Modulated Prior Diffusion for Medical Image Segmentation

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CONTENTS

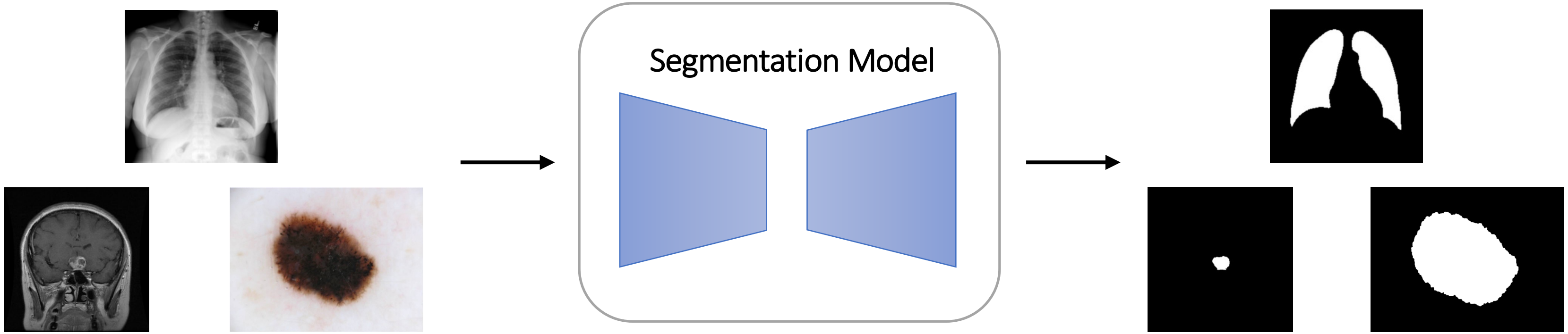
Introduction

Section 1

Section 1 : Introduction

- Background (Image segmentation)

- Medical image segmentation is critical for tasks such as tumor detection, organ delineation, and identifying diseased areas. Since doctors rely on accurate segmentation results to make clinical decisions, it is essential to develop methods that are both precise and computationally efficient. Recent models such as **MedSAM** and **SegDiff** have demonstrated strong performance in this area.



- SAM/MedSAM
- SegDiff

Section 1 : Introduction

- Motivation

- Clinical skin analysis requires accurate segmentation of both facial skin regions and skin lesions.
- Recent methods such as **MedSAM** and **SegDiff** have demonstrated strong segmentation performance.
  - **MedSAM** leverages powerful visual representations learned from large-scale pretraining.
  - **SegDiff** employs diffusion models for high-quality image segmentation.
- However, these approaches often rely on **large and architecturally complex models**, which may limit their practicality in real-world clinical deployment. This motivates us to explore whether more accurate skin region segmentation can be achieved through a more compact and efficient model design.

Task :
Clinical Skin Segmentation

→

Existing Approaches:
MedSAM, SegDiff
(Large models)

→

Towards Higher Accuracy with Compact Models

Section 1 : Introduction

- Challenge & Approach

Challenge

- High segmentation accuracy often relies on large or structurally complex models, making it difficult to simultaneously reduce model size while improving accuracy.

Approach

- We propose Modulated Prior Diffusion (MPD), a diffusion-based segmentation framework that incorporates anatomical guidance by directly modulating the diffusion prior, avoiding explicit conditioning and preserving complete reference information.

Section 1 : Introduction
- Contributions

- We propose a novel conditioning strategy for diffusion-based image segmentation, termed **Modulated Prior Diffusion (MPD)**, which incorporates anatomical guidance by modulating the diffusion prior, without additional conditioning networks.

- We introduce a **condition strength parameter w** to control the trade-off between random noise and conditional information during the diffusion process.

- Through experiments on the ISIC public benchmark and the KMU clinical facial dataset, we show that MPD achieves comparable or better segmentation performance than MedSAM and SegDiff with a more compact model design.

Related works

Section 2

Section 2 : Related works
- Image segmentation

- Recent medical image segmentation research has increasingly explored foundation-model-based and diffusion-based approaches.
- In this work, we select two representative methods, **MedSAM** and **SegDiff,** as our primary baselines to reflect these two research directions.

[1] Ma, J., He, Y., Li, F., Han, L., You, C., & Wang, B. (2024). Segment anything in medical images. Nature Communications, 15(1), 654.

[2] Amit, T., Shaharbany, T., Nachmani, E., & Wolf, L. (2021). Segdiff: Image segmentation with diffusion probabilistic models. arXiv preprint arXiv:2112.00390.

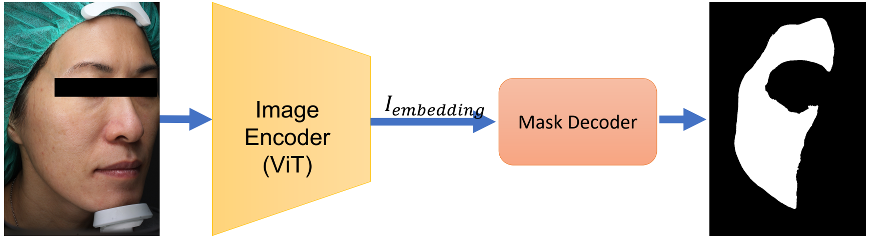
Section 2 : Related works
- MedSAM

Design

- **MedSAM** is a foundation-model-based medical image segmentation method.
- It relies on a **large pre-trained image encoder** to learn general medical visual representations, followed by a decoder to generate segmentation masks.

Limitation

- The segmentation performance is strongly tied to the scale of the backbone, resulting in high memory usage. Such a design may be difficult to deploy in resource-limited clinical settings.



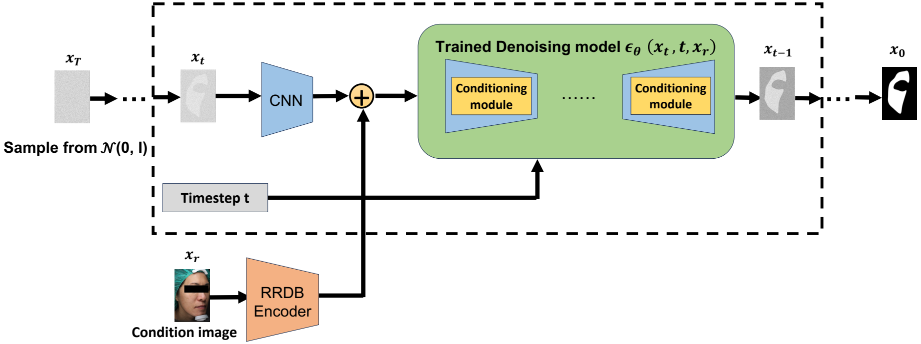
Section 2 : Related works
- SegDiff

Design

- **SegDiff** is a diffusion-based medical image segmentation method.
- It formulates segmentation as a multi-step denoising process, where a noisy segmentation mask is progressively refined.
- At every diffusion step, features of the noisy segmentation are fused with conditional features for denoising.

Limitation

- Because feature fusion is required at every diffusion step and conditional information is processed through a conditioning module, SegDiff involves a more complex inference pipeline and higher inference cost.



Proposed method

Section 3

Section 3 : Proposed methed
- Key formulation of MPD

- Gaussian Prior Conditional Diffusion (GPCD, i.e., traditional conditional diffusion)
  - The reverse process is initialized from a pure Gaussian prior.

$$x_T \sim \mathcal{N}(0, \mathrm{I})$$

  - The reverse denoising process starts from a noisy sample that does not contain conditioning information.

- Modulated Prior Diffusion (MPD)
  - The reverse process is initialized from a modulated prior whose mean is shifted toward the reference image.

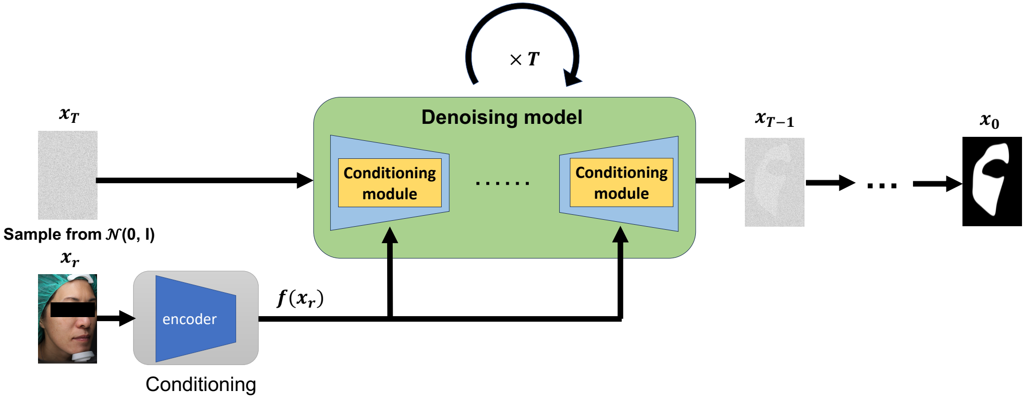
$$x_T \sim \mathcal{N}(x_r, \mathrm{I})$$

  - The reverse denoising process starts from a noisy sample that already contains conditioning information.

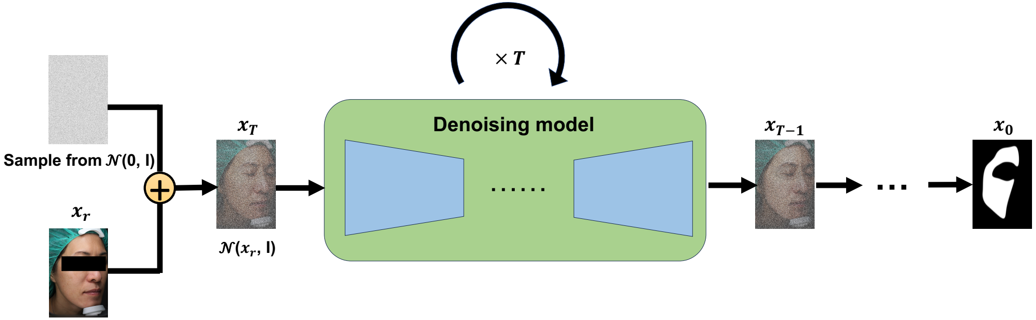
Section 3 : Proposed methed

- Modulated Prior Diffusion (MPD) vs. Gaussian Prior Conditional Diffusion (GPCD)

- Gaussian Prior Conditional Diffusion (GPCD)



- Modulated Prior Diffusion (MPD)

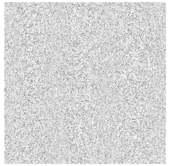


Section 3 : Proposed method
- Condition Strength Parameter (w)

- w ∈ [0,1] controls the strength of the reference image
- (1 − w) controls the noise variance

$$\epsilon \sim \mathcal{N}(w\, x_r\, ,\, (1 - w)\mathrm{I}\,)$$

- Conceptual illustration of condition strength w

 $\sim \; w \; \times$  $+ \; (1 - w) \times$ 

Section 3 : Proposed method
- MPD Forward Process Modification

- MPD Forward Process Equation :

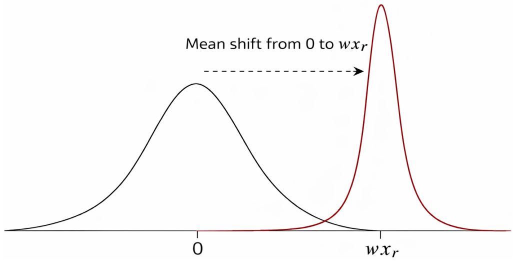
$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon \quad , \quad \epsilon \sim \mathcal{N}(\, w x_r, (1 - w)\, \mathrm{I}\,)$$

- Conceptual illustration of the forward process. Noise ϵ added to x_0 contains reference image information.


$= \sqrt{\bar{\alpha}_t}$

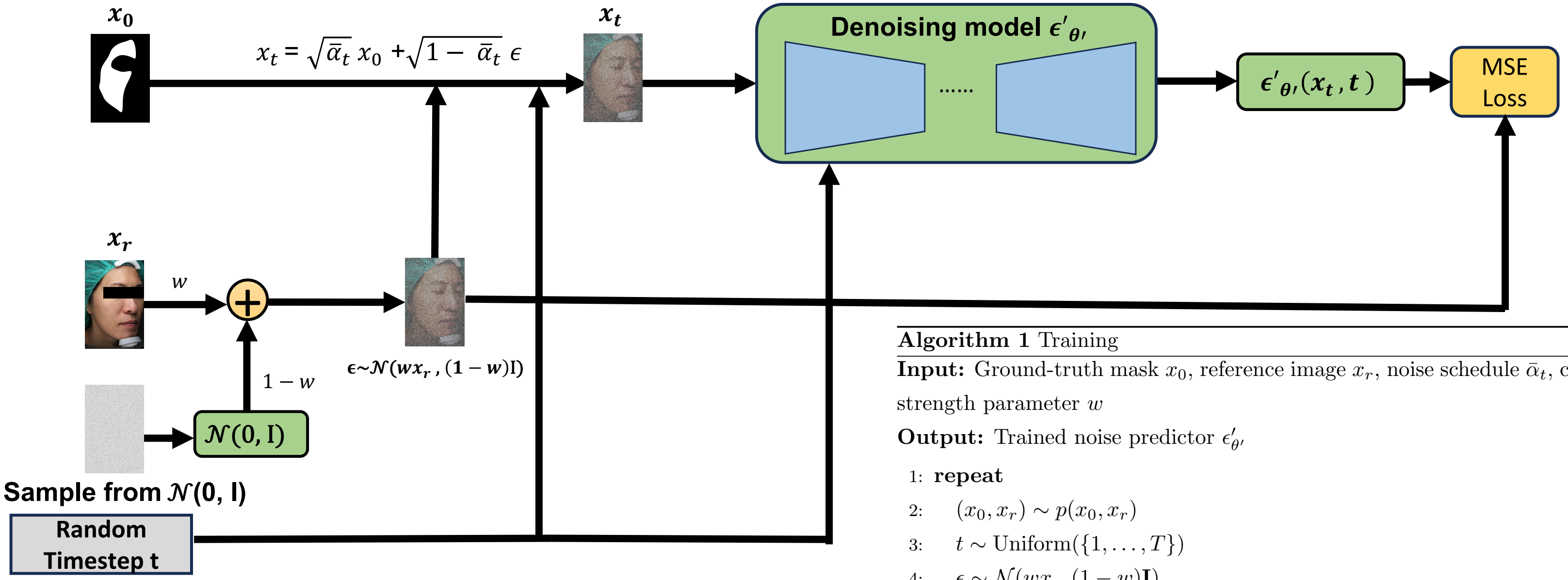
$+ \sqrt{1 - \bar{\alpha}_t}$


- One-dimensional illustration of the noise distribution. The Gaussian mean shifts from 0 to wx_r , and the distribution becomes narrower due to reduced noise variance.



Section 3 : Proposed methed

- Training MPD



Algorithm 1 Training

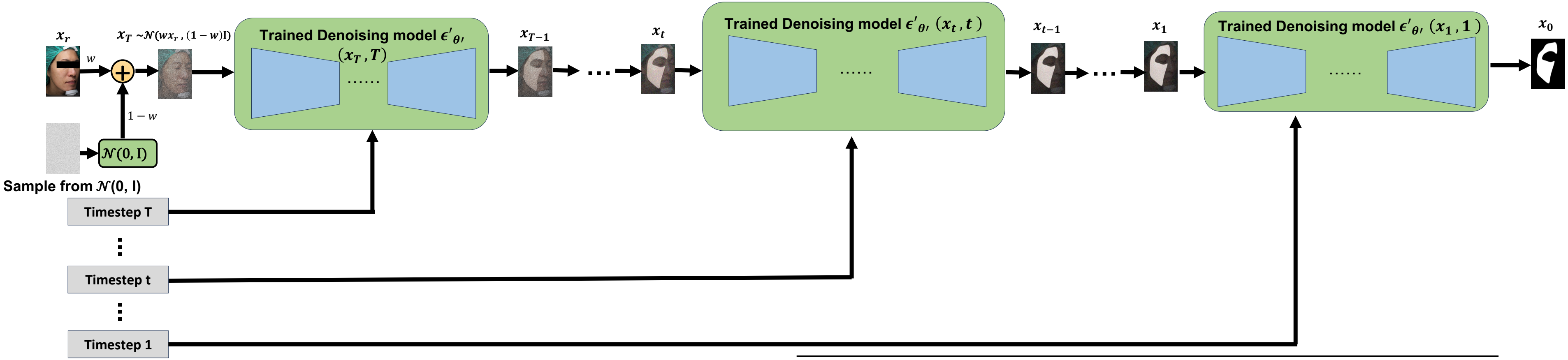
Input: Ground-truth mask x_0, reference image x_r, noise schedule $\bar{\alpha}_t$, condition strength parameter w

Output: Trained noise predictor $\epsilon'_{\theta'}$

1: **repeat**
2: $(x_0, x_r) \sim p(x_0, x_r)$
3: $t \sim \text{Uniform}(\{1, \dots, T\})$
4: $\epsilon \sim \mathcal{N}(wx_r, (1-w)\mathbf{I})$
5: $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$
6: $\hat{\epsilon} \leftarrow \epsilon'_{\theta'}(x_t, t)$
7: $\mathcal{L} \leftarrow \|\epsilon - \hat{\epsilon}\|^2$
8: Update θ' via backpropagation
9: **until** converged

Section 3 : Proposed methed
- Sampling

$$x_{t-1} = \sqrt{\alpha_{t-1}}\left(\frac{x_t-\sqrt{1-\alpha_t}\,\tilde{\epsilon}_t}{\sqrt{\alpha_t}}\right)+\sqrt{1-\alpha_{t-1}}\tilde{\epsilon}_t$$



Algorithm 2 Sampling

Require: x_r : reference image

Require: w : condition strength parameter

1: $x_T \sim \mathcal{N}(wx_r, (1-w)\mathbf{I})$

2: **for** $t = T, \ldots, 1$ **do**

3: $\tilde{\epsilon}_t = \epsilon'_{\theta'}(x_t, t)$

4: $x_{t-1} = \sqrt{\alpha_{t-1}}\left(\frac{\mathbf{x}_t-\sqrt{1-\alpha_t}\tilde{\epsilon}_t}{\sqrt{\alpha_t}}\right) + \sqrt{1-\alpha_{t-1}} \cdot \tilde{\epsilon}_t$

5: **end for**

6: **return** x_0

Section 3 : Proposed method
- Loss Function Comparison

- GPCD Loss Function

$$\mathcal{L}_{GPCD} = \mathbb{E}_{x_0, x_r, \epsilon, t}[||\epsilon - \epsilon_\theta(x_t, f(x_r), t)||^2]$$

- Notation
  - x_0 : ground-truth segmentation mask
  - x_r : reference image
  - ϵ : Gaussian prior
  - $t \sim$ Uniform ({1,…,T})

- In GPCD, the condition is first encoded into a latent representation $f(x_r)$, which may introduce information compression before being fed into the denoising model.

- MPD Loss Function

$$\mathcal{L}_{MPD} = \mathbb{E}_{x_0, x_r, \epsilon, t}[||\epsilon - \epsilon'_{\theta'}(x_t, t)||^2]$$

- Notation
  - x_0 : ground-truth segmentation mask
  - x_r : reference image
  - ϵ : Modulated prior
  - $t \sim$ Uniform ({1,…,T})

- In MPD, the condition modulates the prior mean, so the reference information influences the diffusion process without undergoing encoding or information compression.

Section 3 : Proposed method
- Information-Theoretic Hypothesis

- Hypothesis

Let x_r encode anatomical structure information. Under the same model capacity and training data, we hypothesize that Modulated Prior Diffusion (MPD) is expected to achieve a lower expected training loss than Gaussian Prior Conditional Diffusion (GPCD) in segmentation tasks, due to its ability to preserve more complete conditioning information in the forward process.

$$\mathbb{E}\left[\left\|\epsilon - \epsilon'_{\theta'}{}^{MPD}(x_t, t)\right\|^2\right] \leq \mathbb{E}\,[\|\epsilon - \epsilon_\theta{}^{GPCD}(x_t, f(x_r), t)\|^2]$$

Section 3 : Proposed method
- Hypothesis Rationale – Conditioning Path

- Rationale

Assume a model with finite capacity. Then:

1. Under equivalent training data, x_r in GPCD must pass through a learned transformation $f(x_r)$, subject to compression and information loss.
2. In MPD, x_r is injected directly and linearly into the generative prior, ensuring full access to anatomical structure.
3. By the Data Processing Inequality, the effective mutual information between model input and x_r satisfies

$$\boldsymbol{I}(\boldsymbol{Z}; x_r)_{\boldsymbol{MPD}} \geq \boldsymbol{I}(\boldsymbol{Z}; f(x_r))_{\boldsymbol{GPCD}}$$

meaning MPD preserves higher conditional information.

Section 3 : Proposed method
- Hypothesis Implication – Risk Perspective

- Implication

4. According to the Information–Risk Tradeoff theorem (Poole et al., 2019; Xu & Raginsky, 2017), higher mutual information is associated with a lower achievable risk bound $\boldsymbol{R}^*(\boldsymbol{Z})$:

$$\boldsymbol{R}^*(\boldsymbol{Z})_{\boldsymbol{MPD}} \leq \boldsymbol{R}^*(\boldsymbol{Z})_{\boldsymbol{GPCD}}$$

5. As a result, we expect that the learned function $\epsilon'_{\theta'}{}^{\boldsymbol{MPD}}$ approximates the true noise ϵ with lower variance, yielding a smaller expected training loss.

$$\mathbb{E}\left[\left\|\epsilon - \epsilon'_{\theta'}{}^{MPD}(x_t, t)\right\|^2\right] \leq \mathbb{E}\left[\|\epsilon - \epsilon_\theta{}^{GPCD}(x_t, f(x_r), t)\|^2\right]$$

Thus, MPD minimizes the objective more efficiently.

Experiments & Results

Section 4

Section 4 : Experiments
- Experimental Setup

➢ Task

- Medical image segmentation

➢ Datasets

- **ISIC Dataset** : Public benchmark for generalization
- **KMU Facial Dataset** : Target application dataset

➢ Methods

- **Proposed** : Modulated Prior Diffusion (MPD)
- **Baselines** : MedSAM & SegDiff

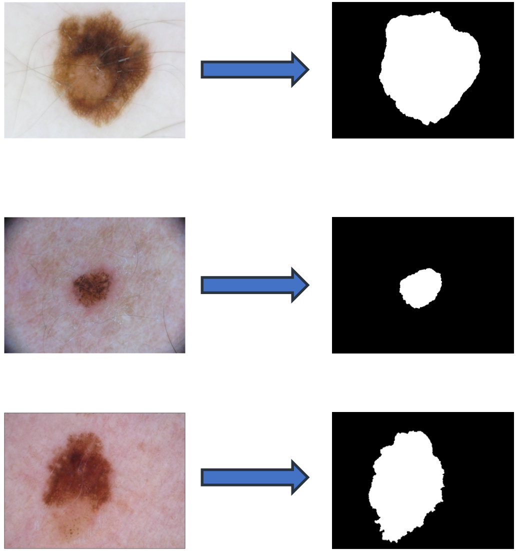
➢ Evaluation

- Intersection over Union (IoU)
- Weighted IoU (for overall cross-dataset evaluation)

Section 4 : Experiments
- Dataset I: Skin Lesion Segmentation (ISIC)

- This study evaluates the proposed method using the ISIC skin lesion dataset.
- The dataset provides ground truth masks annotated by medical experts for skin lesion segmentation.
- The task focuses on segmenting lesion regions from dermoscopic images, where lesion boundaries are often irregular and visually ambiguous.

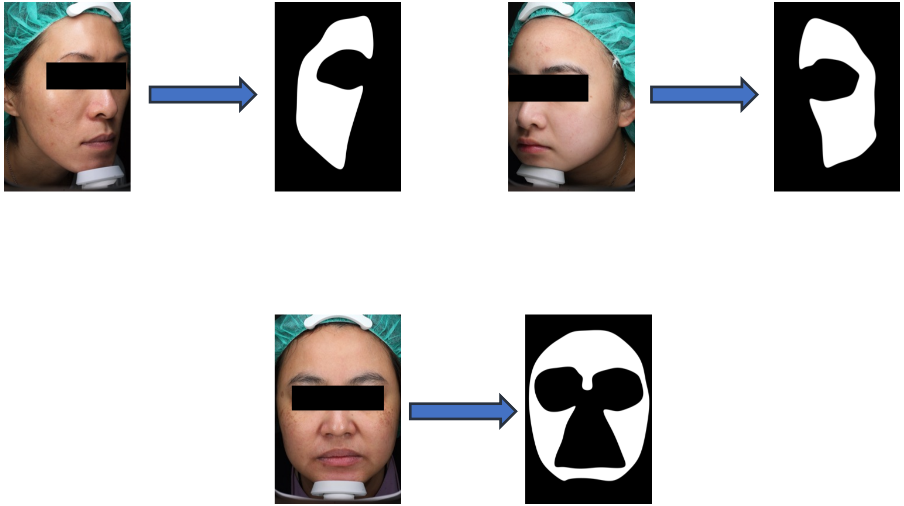
Dataset	
Number of images	1279
Image modality	Dermoscopic images
Resolution	~1020× 760 pixels



Section 4 : Experiments
- Dataset II: Facial Skin Region Segmentation (KMU)

- This study evaluates the proposed method using a clinical facial image dataset collected at Kaohsiung Medical University (KMU).
- All ground truth segmentation masks were manually annotated by medical professionals.
- The annotations focus on clinically relevant facial skin regions for segmentation.

Dataset	
Number of images	249 clinical facial images
View angles	Left, right, frontal
Resolution	~3500 × 5200 pixels
Age range	20–80 years



Section 4 : Experiments
- Observations on the Datasets and Evaluation Considerations

- **The two datasets differ substantially in size.**
  The ISIC dataset serves as a large public benchmark, whereas the KMU dataset is comparatively smaller and clinically collected.
- **Directly averaging the overall results of the two datasets with equal weight may introduce bias.**
  The dataset with fewer samples could have an excessive influence on the final evaluation, thereby affecting the fairness of the comparison.
- **In addition to reporting IoU for each dataset, a dataset-size-weighted overall IoU is also computed.**
  To ensure fairness, the weighting is based on the number of test samples, with an approximate weighting ratio of 8 : 1 between ISIC and KMU.

Section 4 : Experiments
- Compared Methods in Experiments

- The selected methods represent different model categories in medical image segmentation, enabling a structured comparison of MPD with both diffusion-based and non-diffusion baselines.

  - MedSAM
    - Representative large-scale pretrained foundation model
  - SegDiff
    - Representative diffusion-based segmentation model
  - MPD
    - Proposed diffusion-based segmentation method

Section 4 : Experiments
- Evaluation Metric

➢ Intersection over Union (IoU)

- Measures region-level overlap between prediction and ground truth.

$$\mathrm{IoU} = \frac{|P \cap G|}{|P \cup G|}$$

➢ Weighted IoU

- Computes an overall IoU score by weighting dataset-wise IoU according to the number of test samples.

$$\textbf{Weighted IoU} = \frac{N_{\text{ISIC}} \cdot \mathbf{IoU}_{\text{ISIC}} + N_{\text{KMU}} \cdot \mathbf{IoU}_{\text{KMU}}}{N_{\text{ISIC}} + N_{\text{KMU}}}$$

where N_{ISIC} and N_{KMU} are the numbers of test samples in the ISIC and KMU datasets, and $\mathrm{IoU}_{\text{ISIC}}$ and $\mathrm{IoU}_{\text{KMU}}$ are the corresponding average IoU scores.

Section 4 : Results
- Model Size Comparison

- MPD achieves competitive segmentation performance with a relatively compact model size.

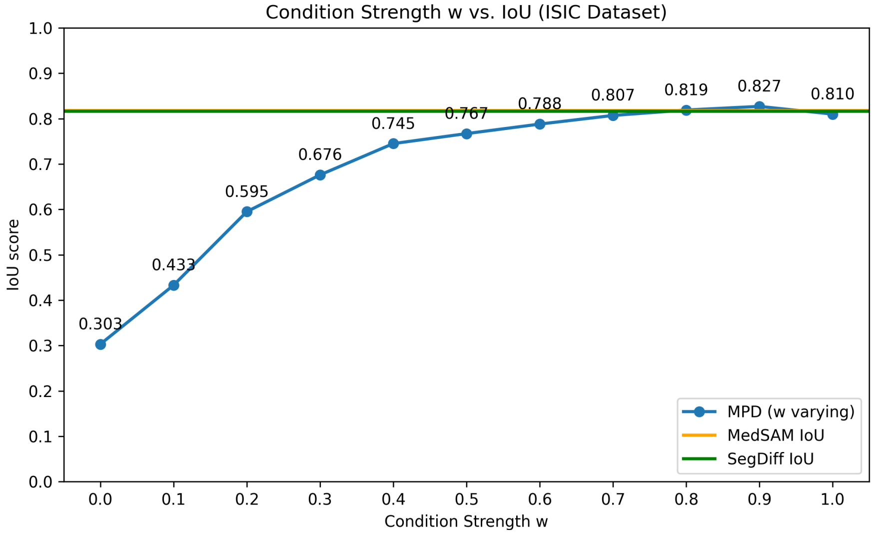
Method	Parameters (M)	Model Size (MB)
MedSAM	90.6	345.7
SegDiff	40.7	155.4
MPD(our method)	**38.5**	**147.0**

Section 4 : Results

- Effect of Condition Strength w on IoU (ISIC Dataset)

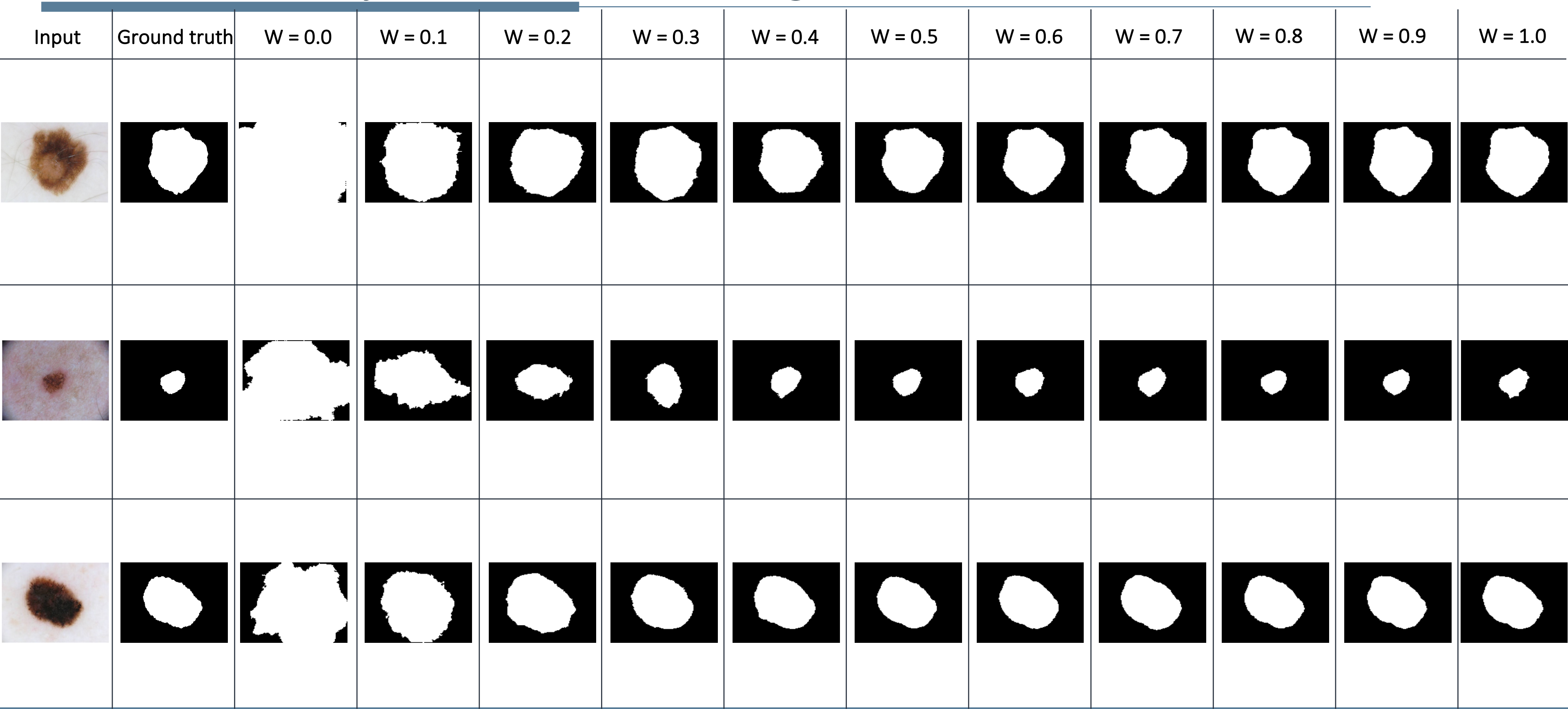
Observation:

On the ISIC dataset, as the condition strength **w** increases, the IoU score improves steadily and reaches its peak at **w = 0.9**. Beyond this point, the performance begins to decline, indicating that further increasing **w** does not lead to continued improvement. At this setting, MPD also outperforms the MedSAM and SegDiff baselines.



Section 4 : Results

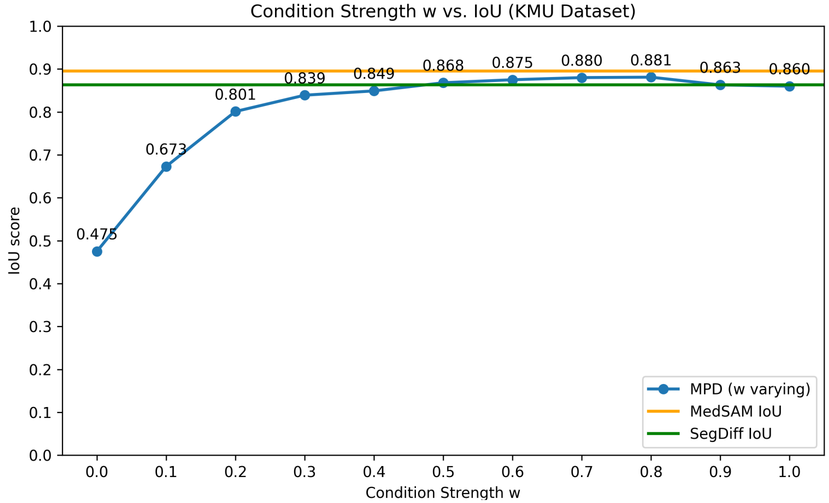
- Qualitative Analysis of Condition Strength w (ISIC Dataset)



Section 4 : Results
- Effect of Condition Strength w on IoU (KMU Dataset)

Observation:

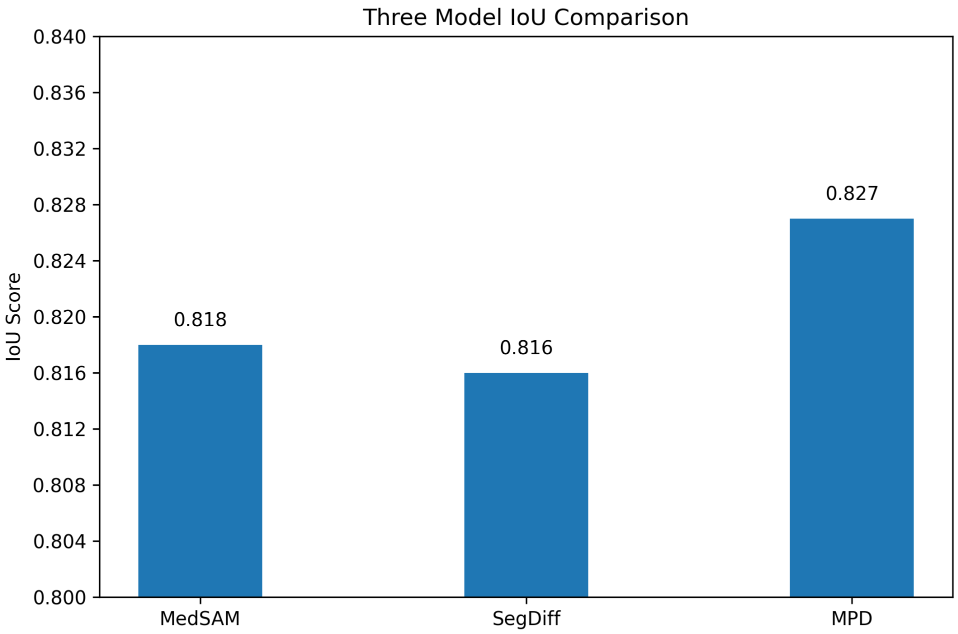
On the KMU dataset, a similar trend is observed, where increasing the condition strength **w** leads to improved IoU performance, with the best result achieved at w = 0.8. When w exceeds this value, the IoU score shows a slight decrease. At this setting, MPD achieves performance comparable to MedSAM and outperforms SegDiff.



Section 4 : Results
- Qualitative Analysis of Condition Strength w (KMU Dataset)

Section 4 : Results
- Comparison with Baseline Methods (ISIC Dataset)

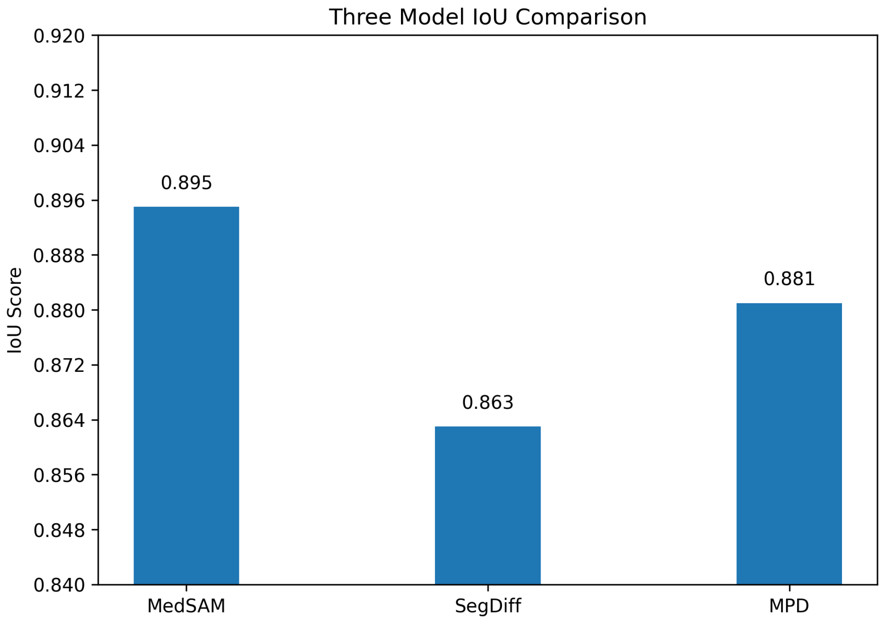
- On the ISIC dataset, MPD achieves the highest IoU among the compared methods.



Part3: Results

- Comparison with Baseline Methods (KMU Dataset)

- On the KMU dataset, MPD demonstrates competitive IoU performance among the compared methods.



Section 4 : Results
- Overall Performance Across Datasets

- The weighted IoU is computed as a weighted average of the ISIC and KMU test sets, based on their dataset sizes.
- MPD maintains competitive performance across both datasets.

Method	ISIC IoU	KMU IoU	Weighted IoU
MedSAM	0.818	**0.895**	0.827
SegDiff	0.816	0.863	0.822
MPD(our method)	**0.827**	0.881	**0.833**

Conclusion

Section 5

Section 5 : Conclusion
- Conclusion

- This work presents **Modulated Prior Diffusion (MPD)**, a conditional diffusion framework for medical image segmentation.
- Instead of introducing additional encoders or complex fusion modules, MPD injects conditional information by **modulating the diffusion prior at an early stage**.
- MPD achieves comparable or better IoU performance on both KMU and ISIC datasets when compared with MedSAM and SegDiff.
- The proposed approach maintains **a compact and efficient architecture**, reducing reliance on large pretrained models or complex network designs.
- The condition strength w provides **controllable conditional guidance**, leading to **improved IoU as w increases until performance saturates**.

Thanks for Your Attention

Reporter：李秋賢 Date ：2026/4/7